

Coal and Petroleum

Coal:

- It is hard and black in colour.

Story of Coal:

- Due to natural processes, the dense forests got buried under the soil.
- Under high temperature and high pressure, dead plants got slowly converted to coal.
- The slow process of conversion of dead vegetation into coal is called carbonisation.

There are different kinds of coals:

Peat: It has a carbon content of 28%.

Lignite: It has 28-30% content of carbon.

Bituminous coal: It comprises of 78-86% carbon.

Anthracite: This is the matured form of coal. It has 94-98 % of carbon.

- **Coal**
- It is used as fuel to cook food. Earlier, it was used in railway engines to produce steam.
- It is used in thermal power stations to generate electricity.
- **Coke, coal tar, and coal gas are the products of coal.**
- **Coke:**
- It is a tough, porous and black substance.
- It is the pure form of carbon and is used in the extraction of steel and many other metals.
- **Coal tar:**
- It is black, thick liquid with unpleasant smell.
- It is the mixture of about 200 substances.
- The products obtained from coal tar are used as starting materials for dyes, drugs, paints, perfumes, etc.
- **Coal gas:**

- It is obtained during the processing of coal to obtain coke.
- It is used as a fuel in many industries situated near the coal processing plants.

Natural Gas

- Natural gas is formed from dead organisms which decompose in the absence of air under conditions of high pressure and temperature.
- It is stored under high pressure as compressed natural gas (CNG). It is used as a fuel for vehicles because it is a cleaner fuel (less polluting).

LPG

- LPG stands for liquified petroleum gas.
- The mixture of these gases is liquified at high pressure and then filled in cylinders.
- It is usually used as a household fuel.
- Coal, petroleum, and natural gas were formed from the dead remains of living organisms. Hence, they are known as **fossil fuels**
- Resources such as coal and petroleum are limited. Burning of such fuels is the major cause of air pollution. Therefore, these fuels should be used only when necessary.
- Burning of fossil fuels releases gases like carbon dioxide which cause greenhouse effect.

Constituents of Petroleum	Uses
LPG	As fuel for home and industry
Petrol	As motor fuel, aviation fuel
Paraffin wax	Candle, vaseline, etc.
Diesel	As fuel for heavy motor vehicles
Kerosene	As fuel for stoves, lamps

- Complete combustion of hydrocarbons leads to formation of carbon dioxide and water, while incomplete combustion of hydrocarbons yields carbon monoxide and water.
- Carbon dioxide as well as carbon monoxide are responsible for pollution in our environment.

- The test used to determine the amount of carbon monoxide in vehicle's exhaust gases is known as emission test.
- Gas analyzer is used in detection of carbon monoxide in exhaust gases.
- **Non-renewable sources** of energy are those that are consumed at a rate faster than that at which they are replenished. Example: Fossil fuels.
- **Fossil fuels** – Coal, petroleum and natural gas
 - **Coal:** It is a non-renewable source of energy made up of complex compounds of carbon, hydrogen and oxygen along with some free carbon and compounds of nitrogen and sulphur.
 - **Petroleum:** It is a dark coloured viscous liquid also known as crude oil or black gold. It is a complex mixture of many hydrocarbons with water, salt, earth particles and other compounds of carbon, oxygen, nitrogen and sulphur. We obtain petroleum by drilling oil wells into earth's crust at its reservoirs. The petroleum extracted from wells has to be purified to obtain different useful components. The process of separating useful components from the crude oil is called refining and this process is done by fractional distillation in big refineries.
 - **Natural Gas:** The main constituents of natural gas are methane (upto to 95%), ethane and propane. It easily burns to produce heat.
- **Advantages –**
 - Easy availability
 - Generate heat that is easily converted into electricity
- **Renewable sources of energy** are those that are replenished at a rate faster than that at which they are consumed. Example:
 - **Solar energy** – Solar cooker, solar water heater (very efficient for small scale electricity production)
 - **Tidal energy, wave energy, ocean thermal energy**
 - **Geothermal energy** – Heat energy inside the earth
 - **Nuclear energy** – Not dependent on solar energy, never-ending source, very efficient source, more environment friendly.
- Coal, petroleum, and natural gas were formed from the dead remains of living organisms. Hence, they are known as **fossil fuels**

- Resources such as coal and petroleum are limited. Burning of such fuels is the major cause of air pollution. Therefore, these fuels should be used only when necessary.
- Burning of fossil fuels releases gases like carbon dioxide which cause greenhouse effect.

Coal:

- It is hard and black in colour.

Story of Coal:

- Due to natural processes, the dense forests got buried under the soil.
- Under high temperature and high pressure, dead plants got slowly converted to coal.
- The slow process of conversion of dead vegetation into coal is called carbonisation.

There are different kinds of coals:

Peat: It has a carbon content of 28%.

Lignite: It has 28-30% content of carbon.

Bituminous coal: It comprises of 78-86% carbon.

Anthracite: This is the matured form of coal. It has 94-98 % of carbon.

- **Coal**
- It is used as fuel to cook food. Earlier, it was used in railway engines to produce steam.
- It is used in thermal power stations to generate electricity.
- **Coke, coal tar, and coal gas are the products of coal.**
- **Coke:**
- It is a tough, porous and black substance.
- It is the pure form of carbon and is used in the extraction of steel and many other metals.
- **Coal tar:**
- It is black, thick liquid with unpleasant smell.
- It is the mixture of about 200 substances.
- The products obtained from coal tar are used as starting materials for dyes, drugs, paints, perfumes, etc.
- **Coal gas:**

- It is obtained during the processing of coal to obtain coke.
- It is used as a fuel in many industries situated near the coal processing plants.

Petroleum:

- Petroleum was formed from the dead organisms present in the sea.
- Petrol, diesel, kerosene, paraffin wax, lubricating oil, and petroleum gas are the products of petroleum.
- The process of separating various constituents of petroleum is known as **refining**.
- Refining of petroleum is done in fractionating column.
 - Components with higher boiling points are collected at the bottom of the fractionating column.
 - Components with lower boiling points are collected at the top of the column.

Cracking

- Heating higher alkanes to sufficiently high temperatures in absence of oxygen, in order to obtain lower hydrocarbons is known as cracking or pyrolysis.

Constituents of Petroleum	Uses
LPG	As fuel for home and industry
Petrol	As motor fuel, aviation fuel
Paraffin wax	Candle, vaseline, etc.
Diesel	As fuel for heavy motor vehicles
Kerosene	As fuel for stoves, lamps

- Complete combustion of hydrocarbons leads to formation of carbon dioxide and water, while incomplete combustion of hydrocarbons yields carbon monoxide and water.
- Carbon dioxide as well as carbon monoxide are responsible for pollution in our environment.

- The test used to determine the amount of carbon monoxide in vehicle's exhaust gases is known as emission test.
- Gas analyzer is used in detection of carbon monoxide in exhaust gases.
- is Petroleum a natural resource
- The formation of petroleum takes place over a period of millions of years.
- Petroleum reserves are limited i.e. they are found only in a few places on the Earth.
- The ever increasing population has further increased the demand for petroleum and its products.
- In India, the Petroleum Conservation Research Association (PCRA) advises people on the various methods that can be adopted to conserve petrol/diesel.

Natural Gas

- Natural gas is formed from dead organisms which decompose in the absence of air under conditions of high pressure and temperature.
- It is stored under high pressure as compressed natural gas (CNG). It is used as a fuel for vehicles because it is a cleaner fuel (less polluting).

LPG

- LPG stands for liquified petroleum gas.
- The mixture of these gases is liquified at high pressure and then filled in cylinders.
- It is usually used as a household fuel.